



MAGALANG CHRISTIAN ECUMENICAL SCHOOL, INC.
Marbea Subd., Sta. Cruz, Magalang, Pampanga

METHODS

Research Design

This study will employ a quantitative research design because it enables the researchers to collect numerical data from a sufficiently large population and perform statistical analyses that reveal patterns and relationships among variables. This approach is appropriate because it allows the researchers to examine the relationship between the utilization of ChatGPT and QuillBot among Grade 12 students and their academic performance. Quantitative research also helps reduce the influence of bias through systematic data collection and analysis, enabling the researchers to accurately interpret the data and produce valid and reliable findings (Alford & Teater, 2025).

Specifically, the study will employ a non-experimental correlational research design to examine the relationship between the utilization of ChatGPT and QuillBot among Grade 12 students and their academic performance. According to Sreekumar (2026), correlational research allows researchers to examine the relationship between two or more variables without manipulating or controlling them. Guided by this design, the researchers will gather data on students' utilization of ChatGPT and QuillBot, the independent variable, and their academic performance, the dependent variable, and will then analyze the data collected to determine whether a statistically significant relationship exists between the two, consistent with the hypotheses stated in Chapter 1. Furthermore, this design is suitable because it allows the researchers to examine the relationship between the utilization of ChatGPT and QuillBot and students' academic performance under natural conditions, without manipulating or controlling either variable.

Respondents/Participants

The respondents of this study will consist of 50 Grade 12 students enrolled at Magalang Christian Ecumenical School, Inc. (MCESI) during the School Year 2026–2027. These respondents will be selected from the population of eligible Grade 12 students who meet the inclusion criteria established by the researchers. A sample of 50 respondents is considered sufficient to provide adequate quantitative data for examining the relationship between the utilization of ChatGPT and QuillBot and academic performance while remaining manageable within the scope, time frame, and resources of the study. A simple random sampling technique will be employed to ensure that every eligible Grade 12 student has an equal chance



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of being selected to participate in the study. This sampling method was chosen because it minimizes selection bias and allows the sample to reasonably represent the Grade 12 student population of MCESI.

Instruments

Data for this study will be collected using a researcher-made questionnaire designed to measure Grade 12 students' utilization of ChatGPT and QuillBot and their academic performance, consistent with the variables identified in the conceptual framework in Chapter 1. The questionnaire will be composed of two main parts. Part I will measure students' utilization of ChatGPT and QuillBot in terms of frequency of AI tool usage, academic purposes of AI usage, dependence on AI assistance, and integration into study routines using Likert-scale items. Part II will gather respondents' final general average from the previous academic year, which will serve as the measure of their academic performance.

To ensure content validity, the instrument will be reviewed by three experts consisting of the research adviser, a Statistics teacher, and an English teacher to evaluate the questionnaire's content validity, clarity, organization, and appropriateness for the objectives of the study. Their feedback will guide revisions to the clarity, structure, and content of the items to better align them with the study's objectives. Following content validation, the questionnaire will be pilot-tested among Grade 12 students who are not included in the actual respondents to determine the clarity, consistency, and reliability of the instrument. The pilot-test responses will be analyzed using Cronbach's Alpha to determine the questionnaire's internal consistency reliability. Cronbach's Alpha measures whether the items in the questionnaire consistently measure the same construct. Based on the guide, a Cronbach's Alpha coefficient of 0.70–0.79 indicates acceptable reliability, 0.80–0.89 indicates good reliability, and 0.90–1.00 indicates excellent reliability. If the computed Cronbach's Alpha is 0.70 or higher, the questionnaire will be considered sufficiently reliable for use in the actual data collection. Otherwise, the instrument will be revised based on the pilot-test results before it is administered to the respondents.

Ethical Considerations

Ethical considerations will be strictly observed throughout the conduct of this study. Before participating, each respondent will be informed of the study's purpose, what their participation will involve, and how their responses will be used. Respondents will be asked to provide informed consent prior to answering the questionnaire, and participation will be entirely voluntary; no student will be required or pressured to take part or to answer any item they are uncomfortable with.



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All personal information collected will be kept private and confidential. Any details that could identify individual respondents will be removed from the data, and the information gathered will be used solely for the purposes of this research and will not be shared with anyone outside the research group. These measures are intended to protect respondents' privacy, safeguard the confidentiality of their information, and ensure that the study follows accepted ethical research guidelines. The collected data will be stored in a password-protected digital file accessible only to the researchers and research adviser. All data will be retained solely for the completion of the study and will be securely deleted after the research has been completed.

Data Collection

The researchers will first secure permission from the school administration to conduct this study. Once approval is obtained and the instrument has been validated and pilot-tested as described above, the finalized questionnaire will be administered through a Google Form distributed to the selected Grade 12 respondents through the school's online platforms.

Before answering the questionnaire, respondents will be given an informed consent form explaining the purpose of the study, the voluntary nature of their participation, and the confidentiality and anonymity of their responses. Respondents will confirm their consent by reading the form and checking the corresponding checkbox before proceeding to the questionnaire. Completing the survey is expected to take about 10 to 15 minutes, and the researchers will be available online to answer questions and clarify instructions as needed. Follow-up reminders will be sent when necessary to improve the response rate and ensure the completeness of the data.

Once all responses have been collected, the researchers will review the completed questionnaires for completeness, consistency, and accuracy. The responses will then be encoded into Microsoft Excel, checked for missing or erroneous entries, and prepared for statistical analysis using Jamovi statistical software. The software will be used to compute the descriptive and inferential statistical measures required to answer the research questions and test the hypotheses.

Data Analysis/Statistical Treatment of Data

The data gathered through the questionnaire will be analyzed using both descriptive and inferential statistical methods through Jamovi statistical software. Each statistical tool will be selected according to the specific research question it is intended to answer.



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Weighted Mean. The weighted mean will be used to determine the extent of Grade 12 students' utilization of ChatGPT and QuillBot in terms of frequency of AI tool usage, academic purposes of AI usage, dependence on AI assistance, and integration into study routines. It will summarize the respondents' level of utilization based on their responses to the Likert-scale questionnaire items.

Mean. The mean will be used to determine the average academic performance of the respondents using their final general average from the previous academic year. As a measure of central tendency, it will summarize the respondents' overall academic achievement. The computed mean will be interpreted using the school's grading scale to describe the respondents' overall level of academic performance.

Pearson Product-Moment Correlation Coefficient (Pearson's r). The Pearson Product-Moment Correlation Coefficient (Pearson's r) will be used to determine the strength, direction, and statistical significance of the relationship between the utilization of ChatGPT and QuillBot and the academic performance of Grade 12 students. This statistical test is appropriate because the study employs a quantitative correlational research design and seeks to determine whether a significant relationship exists between the independent and dependent variables. The significance level will be set at 0.05. If the computed p-value is less than or equal to 0.05, the null hypothesis will be rejected, indicating that a statistically significant relationship exists between the variables. Conversely, if the computed p-value is greater than 0.05, the null hypothesis will fail to be rejected, indicating that no statistically significant relationship exists between the variables.



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REFERENCES

- Alford, S., & Teater, B. (2025). Quantitative research. In Handbook of research methods in social work (pp. 156–171). Edward Elgar Publishing. <https://doi.org/10.4337/9781035310173.00023>
- Sreekumar, D. (2026, June 2). What is correlational research: Definition, types, and examples. Researcher.Life.
<https://researcher.life/blog/article/what-is-correlational-research-definition-and-examples/>